Comprehensive Road Scene Understanding for Autonomous Driving

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Abstract

The present work analyzes the application of anomaly segmentation techniques, focusing on the mask-based EoMT architecture. Three different checkpoints of the model were compared: a COCO-pretrained model, a Cityscapes-pretrained model, and a model obtained by fine-tuning the COCO-pretrained EoMT on the Cityscapes dataset. To guarantee a fair quantitative comparison between the two pre-trained models, a COCO-to-Cityscapes class mapping was applied. The fine-tuning was performed through a progressive layer unfreezing strategy. The best fine-tuned configuration reached 75.2% mIoU on the Cityscapes validation set, a substantial gain over the COCO-pretrained baseline (54.52%), though a gap remains with respect to the Cityscapes-pretrained model (81.68%). The evaluation on the anomaly detection task was then conducted using multiple anomaly scoring post-hoc methods on the Fishyscapes and SegmentMeIfYouCan benchmarks. The Cityscapes-pretrained model proved the most balanced checkpoint, while the fine-tuned model showed degraded anomaly detection performance on some benchmarks, highlighting a trade-off between domain specialization and OoD generalization. Temperature Scaling yielded only marginal improvements, suggesting well-calibrated predictions. The code is available at https://github.com/filoppos/MaskArchitectureAnomaly_CourseProject.

1. Introduction

Autonomous driving systems require accurate and robust scene understanding in order to operate safely in complex and unpredictable environments. Anomaly segmentation represents a key task in this context, as it requires both semantic comprehension and the ability to identify potentially dangerous Out-of-Distribution (OoD) objects in road scenes. Specifically, the joint task of semantic and anomaly segmentation expands the traditional closed-set classification, requiring the model to simultaneously assign predefined class labels to known pixels and report unknown regions as anomalies. To address this, the present paper proposes to evaluate and compare pixel-based and mask-based approaches used for the identification of anomalous regions in real-world scenarios.

Among the most efficient semantic segmentation models for real-time applications, ERFNet introduces an architecture based on residual connections and factorized convolutions, achieving a trade-off between accuracy and computational efficiency [12]. Its simple encoder-decoder architecture allows it to run in real-time on embedded systems. This makes it a solid pixel-wise baseline for comparing mask-based models. However, semantic segmentation alone fails to distinguish different instances of the same class. To overcome this limitation, panoptic segmentation [8] combines semantic and instance information into a unified representation, enabling a more complete understanding of complex scenes.

Alongside these developments, the limitations of traditional pixel-wise semantic segmentation models, such as ERFNet, emerged leading to the development of mask-based models. In particular, MaskFormer redefines the segmentation as a mask classification problem [3], while Mask2Former introduces masked attention, providing greater flexibility across segmentation tasks [4]. Furthermore, the introduction of foundation models such as DINOv2 [11] enabled the learning of more robust and generalizable visual representations, particularly relevant in anomaly segmentation scenario. In this context, Encoder-only Mask Transformer (EoMT) [7] has gained increasing attention, being able to include queries directly into the Vision Transformer and reduce computational costs using the masked attention only at the beginning of training.

To evaluate this model's behaviour in road scenarios, two different EoMT configurations are tested and compared: one trained for semantic segmentation on Cityscapes [5], a road-specific dataset, and the other for panoptic segmen-

tation on COCO [9], a more general-purpose dataset. In order to address this domain mismatch, a specific class-mapping and inference strategy was adopted. Furthermore, the COCO model was fine-tuned on Cityscapes and added to the comparison to investigate whether the performance gap with the domain-specific model can be reduced through adaptation. The ERFNet model was used as a pixel-wise baseline for evaluation.

Anomaly detection benchmarks such as Fishyscapes [1] and SegmentMeIfYouCan [2] were introduced to evaluate the capability of models to detect anomalies and obstacles in autonomous driving scenes, overcoming the limitations of benchmarks focused only on known classes. Evaluation on these datasets highlighted the limitations of traditional Out-of-Distribution detection methods based on softmax confidence, motivating the development of alternative post-hoc anomaly scoring approaches [6, 10], which are evaluated in this work alongside Temperature Scaling for calibration.

2. Methodology

The experimental pipeline is organized into four phases: a comparison between the COCO-pretrained EoMT model and the Cityscapes-pretrained EoMT model for semantic segmentation through a shared class space, a fine-tuning of the COCO-pretrained EoMT model on Cityscapes, an evaluation of anomaly segmentation using post-hoc methods applied to both the pixel-based ERFNet model and the mask-based EoMT model.

2.1. Class Space Mapping

Since the COCO-pretrained EoMT model predicts 133 panoptic classes, whereas the Cityscapes-pretrained model uses only 19 classes, a fair comparison requires a shared semantic space. An initial evaluation performed without any class mapping (Figure 1) revealed a direct incompatibility between the two label spaces. To reduce the loss of spatial details in the high-resolution Cityscapes images, the inference pipeline of the COCO-pretrained model was also converted from a panoptic configuration based on resizing and padding to a semantic configuration using a *sliding window* approach, already adopted by the Cityscapes-pretrained model.

Initially, the class was selected by applying the argmax directly to the COCO class scores, and the resulting prediction was then mapped to the corresponding Cityscapes class:

$$c^*(p) = \arg\max_{c \in C_{\text{COCO}}} s_c(p), \qquad \hat{y}(p) = m(c^*(p)) \quad (1)$$

where $s_c(p)$ denotes the score assigned to class c for pixel p, and $m(\cdot)$ represents the mapping function from COCO classes to Cityscapes classes.

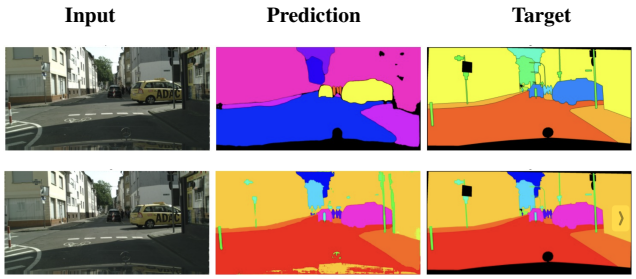


Figure 1. Qualitative comparison between the COCO-pretrained (top) and Cityscapes-pretrained (bottom) models

The final approach instead aggregates the scores of all COCO classes associated with the same Cityscapes class before the final decision:

$$S_k(p) = \sum_{c \in M(k)} s_c(p), \qquad \hat{y}(p) = \arg\max_{k \in C_{\text{CS}}} S_k(p) \quad (2)$$

where $M(k)$ denotes the set of COCO classes mapped to the Cityscapes class k. This aggregation preserves the semantic contribution of related classes and reduces the loss of information caused by the direct conversion between the two label spaces.

2.2. Fine-tuning

Since the mapping alone did not provide sufficiently accurate performance, the COCO-pretrained model was fine-tuned on Cityscapes for the semantic segmentation task. A new model was constructed with a 19-class classification head; pre-trained COCO weights were loaded for all compatible layers, while the classification head was initialized randomly.

To identify the best trade-off between adaptability and knowledge preservation, a *progressive unfreezing* strategy was adopted, consisting of four configurations with an increasing number of trainable parameters: `head-only` (classification and mask heads only), `head-upscale` (added upscaling layers for mask refinement), `head-upscale-query` (added decoder queries), and `Config-blocks-10-11`, in which the last two Transformer blocks of the encoder and the final LayerNorm were also unfrozen. The remaining encoder blocks remained frozen. In contrast, the positional embeddings were kept trainable in all configurations, since their interpolation to the Cityscapes resolution may alter the representations learned during pre-training. The last configuration adopts a discriminative *learning rate*: it starts at 10^{-4} for the task-specific modules and 10^{-5} for the last DINOv2 encoder blocks and the final LayerNorm, thereby reducing the risk of *catastrophic forgetting*. The optimization process uses AdamW (*weight decay* 0.05) with a Cosine An-

nealing scheduler decaying to 10^{-6} over 10 epochs, *Automatic Mixed Precision* (AMP) to reduce training time, *gradient clipping* with a maximum norm of 1.0, and a *batch size* of 2. Each configuration was evaluated using mIoU on the Cityscapes *validation set*, adopting the same evaluation pipeline used in the previous experiments to ensure a fair comparison.

2.3. Post-hoc anomaly scoring methods

The first baseline is the pixel-based ERFNet model. Three anomaly detection methods were evaluated: MSP, which uses the maximum softmax probability as a confidence measure; MaxLogit, which operates directly on the raw logits, avoiding the softmax normalization; and MaxEntropy, which measures uncertainty through the entropy of the softmax distribution.

The evaluation then shifted to the mask-based EoMT model. Since it is not pixel-based, its output is converted into a per-pixel logit map by combining the masks produced by the queries with their corresponding class probabilities:

$$\ell_c(x) = \sum_i p_i(c) \cdot m_i(x) \tag{3}$$

where $m_i(x)$ is the sigmoid mask associated with query i, and $p_i(c)$ is the corresponding class probability.

Each image is resized by scaling its shorter side to the native resolution of the corresponding *checkpoint* (1024 for Cityscapes and 640 for COCO) and processed using a sliding-window semantic inference strategy. For the COCO *checkpoint*, the 133 per-pixel class scores are projected onto the 19 Cityscapes classes through the same mapping function $M(k)$ described in Section 2.1 before applying any anomaly scoring method. This prevents anomalies from being assigned high confidence scores for known panoptic classes. This C-channel representation allows the direct reuse of MSP, MaxLogit, and MaxEntropy without any modification. In addition, RbA (*Rejected by All*) [10] is introduced. Unlike the previous methods, it is specifically designed for mask-based architectures and measures how anomalous the regions rejected by all known classes are. It evaluates the overall agreement of all semantic classes, assigning higher anomaly scores to pixels that are poorly explained by the known categories.

The evaluation is conducted on five benchmarks: SMIYC RoadAnomaly21 (RA-21), SMIYC RoadObstacle21 (RO-21), Fishyscapes Lost&Found (FS L&F), Fishyscapes Static (FS Static), and RoadAnomaly. Three EoMT *checkpoints* are considered: the Cityscapes-pretrained model, the COCO-pretrained model, and the COCO-pretrained model after fine-tuning on Cityscapes.

To ensure a consistent comparison and avoid the memory bottleneck associated with high-resolution multi-channel logit maps, both the *ground-truth* masks and the logit maps are aligned to a fixed evaluation grid of 512×1024 (nearest-neighbor interpolation for the ground truth and bilinear interpolation for the logits). *Void* pixels are excluded, and images without anomalies are discarded. For efficiency, only a single *forward pass* was performed for each image, and the resulting per-pixel logits were stored on disk, allowing them to be reused for all scoring methods and for the entire temperature sweep without re-running the model.

The effect of *Temperature Scaling* is finally analyzed. Since MSP and MaxEntropy depend on softmax probabilities, they are sensitive to the temperature parameter, whereas MaxLogit and RbA, operating directly on the logits, remain unchanged. A temperature *sweep* was therefore performed over different values, selecting for each *checkpoint* the one that maximizes the average AuPRC (Area under the Precision-Recall Curve) on the validation datasets.

2.4. Evaluation Metrics

The quality of semantic segmentation is measured using the *mean Intersection over Union* (mIoU), defined as the average IoU across all classes. Anomaly segmentation is evaluated using two standard metrics, treating anomalies as the positive class: AuPRC, which is robust to the imbalance between normal and anomalous pixels, and FPR95, defined as the false positive rate when the true positive rate reaches 95%.

3. Results

In this section, the results obtained during the experimental work described previously are presented. Table 1 compares the EoMT model pre-trained on Cityscapes with the one pre-trained on COCO, evaluated after the mapping between the two datasets, the sliding-window approach and incorporating an aggregation procedure for COCO class scores before the final label assignment stage. Although these modifications enabled the model to reach a better performance as shown in Figure 2 and an mIoU of 54.52%, the Cityscapes-pretrained model achieves substantially higher performance with an mIoU of 81.68%. Table 1 shows that this gap is driven by classes without a direct correspondence in COCO, such as pole (0.0009) and rider (0.0000).

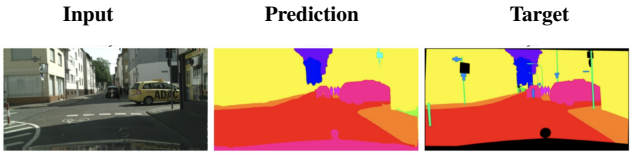


Figure 2. Qualitative results of the COCO-pretrained model after applying the class mapping to Cityscapes.

The application of fine-tuning strategies allowed for a further improvement in mIoU, as shown in Figure 3: the fourth configuration (Config-blocks-10-11) improved the

Table 1. Comparison of class-wise IoU scores achieved by the COCO-pretrained EoMT model and the Cityscapes-pretrained EoMT model after applying the class mapping.

Class	COCO EoMT	Cityscapes EoMT
road	0.9399	0.9840
sidewalk	0.6466	0.8736
building	0.8653	0.9415
wall	0.4578	0.6607
fence	0.4739	0.6549
pole	**0.0009**	**0.7104**
traffic light	0.5948	0.7500
traffic sign	0.0412	0.8213
vegetation	0.8704	0.9302
terrain	0.5436	0.6660
sky	0.8985	0.9554
person	0.6941	0.8538
rider	**0.0000**	**0.7111**
car	0.8411	0.9554
truck	0.2928	0.8179
bus	0.7112	0.9032
train	0.1356	0.7735
motorcycle	0.6300	0.7435
bicycle	0.7210	0.8127
mIoU	**0.5452**	**0.8168**

mIoU from 54.52% to 75.2%, representing the best result among the four configurations considered.

Table 2 reports the comparison of different anomaly scoring strategies applied to the ERFNet and EoMT models. The evaluation is performed on 5 benchmarks (SMIYC RA-21 and RO-21, Fishyscapes L&F and Static, and Road Anomaly) using the AuPRC and FPR95 metrics.

The three checkpoints show complementary behaviours. Cityscapes pre-trained model is the most reliable on road benchmarks, with the best AuPRC on RO-21 (up to 93.45%) and FPR95 close to zero, lowest FPR95 values on FS L&F and Road Anomaly, but modest AuPRC on FS Static (roughly 57%). COCO pre-trained model shows an opposite behaviour: highest AuPRC on RA-21 (up to 72%) and FS Static (roughly 87%) but very high FPR95 on FS L&F (roughly 89%) and Road Anomaly (roughly 78%). Fine-tuned model is intermediate: good on RO-21 and FS Static but critical on RA-21, where the FPR95 increases to 98.3% despite the reasonable AuPRC (roughly 41%), assigning high confidence to the anomalies.

The three methods (MSP, MaxLogit and MaxEntropy) give very similar results (1-2% of variation), while RbA is less stable, despite still being competitive on RO-21 and Road Anomaly.

The impact of the calibration was then evaluated on this model and, as shown in Table 3 , the results remain almost unchanged with respect to the original configuration.This suggests that the application of calibration has a limited effect on the overall performance of the model.

4. Discussion

The obtained results highlight the significant role of the pre-training domain. Among the three EoMT checkpoints the behavior is complementary: COCO pre-trained model has the best AuPRC performances on RA-21 and FS Static, while the Cityscapes pre-trained model is the most balanced generally, combining the highest mIoU with stable anomaly detection with softmax-based scores. The gap does not depend only on the difference between semantic spaces but also on the quality of the representations: being entirely pre-trained on road scenes, the Cityscapes pre-trained model learns more specialized features for the autonomous driving domain.

The fine-tuning activity demonstrates that the COCO-pretrained EoMT model can be successfully adapted to a new domain. However, the fact that its performance does not reach that of the Cityscapes-pretrained EoMT model suggests that some domain-specific information related to road scenes is learned more effectively during pre-training than during the subsequent adaptation phase. This confirms the limitations of transferring knowledge from a general-purpose dataset to a highly specialized domain. Mask-based EoMT also significantly overtakes pixel-based ERFNet baseline, confirming the advantage of mask-based formulation for scoring anomalies.

Something that deserves attention is that AuPRC and FPR95 do not always agree: some configurations have a reasonable AuPRC but a FPR95 close to 100%. This happens because the model assigns similar scores to anomalies and known objects: to identify the 95% of the anomalies, the threshold must be so low that it also flags many normal pixels, causing false positives to explode. The same problem, compounded by the fact the FS L&F objects are very small, explains why every method fails on that dataset.

Among the scoring methods, MSP, MaxLogit and MaxEntropy give results that are very close together while RbA is highly variable: it is the best method on some benchmarks but it is catastrophic on FPR95 of RA-21 and FS Static. Although it is designed for mask architectures, it does not offer a reliable advantage over softmax-based scores. Temperature scaling leaves the metrics almost unchanged: calibration is not sufficient to improve the anomalies detection, which is a sign that the main limitation is not overconfidence but the difficulty in distinguishing true OoD objects from known classes. These results should be interpreted with caution because some test sets are very small and because reducing COCO's 133 classes to 19 loses information, putting the COCO model at a disadvantage in the comparison.

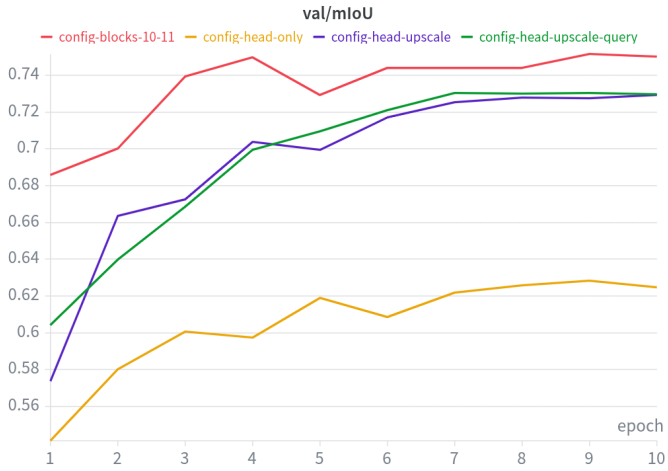


Figure 3. Mean Intersection over Union (mIoU) achieved by the four fine-tuning methods. Config-blocks-10-11: 0.752, Config-head-only:0.6281, Config-head-upscale:0.729, Config-head-upscale-query: 0.7305

Table 2. Comparison of anomaly scoring methods on ERFNet and EoMT models

ERFNet mIoU (69.7%) is the semantic segmentation performance reported in the original ERFNet paper and is provided for reference.

Model	**mIoU**	**Method**	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		RoadAnomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
ERFNet	69.7%	MSP	29.10	62.55	2.71	65.22	1.75	50.60	7.47	41.84	12.42	82.58
		MaxLogit	38.32	59.34	4.63	48.44	3.30	45.49	9.50	40.30	15.58	73.25
		MaxEntropy	30.97	62.66	3.04	65.91	2.58	50.16	8.84	41.55	12.67	82.75
EoMT fine-tuned	75.2%	MSP	41.19	98.32	88.46	0.21	22.25	22.69	64.89	14.94	60.15	20.08
		MaxLogit	40.93	98.37	87.54	0.21	21.85	26.50	64.67	14.39	60.05	19.51
		MaxEntropy	41.24	98.34	87.39	0.20	21.82	22.79	65.17	14.16	60.78	19.01
		RbA	36.57	98.08	73.21	0.43	18.30	85.71	61.25	82.97	58.41	14.63
EoMT COCO	54.5%	MSP	70.57	52.20	90.04	8.09	12.22	89.00	86.80	10.52	71.41	78.70
		MaxLogit	71.36	45.72	90.65	2.98	12.61	89.39	86.94	10.93	71.27	77.93
		MaxEntropy	71.99	33.73	91.42	6.98	13.10	89.20	87.03	9.89	71.13	76.68
		RbA	55.47	67.42	87.36	45.94	9.96	93.99	85.87	51.40	69.75	70.31
EoMT Cityscapes	81.68%	MSP	68.65	29.94	93.18	0.66	16.37	13.00	57.95	41.47	71.46	15.41
		MaxLogit	68.07	32.68	93.22	0.61	16.34	12.80	58.05	47.32	70.90	15.03
		MaxEntropy	68.30	30.37	93.26	0.63	18.68	12.82	56.60	42.96	73.74	14.64
		RbA	64.96	97.43	93.45	0.42	16.42	10.26	60.02	78.86	70.25	14.39

Table 3. Performance comparison of EoMT Cityscapes using MSP with different temperature values.

Method	**SMIYC RA-21**		**SMIYC RO-21**		**FS L&F**		**FS Static**		**Road Anomaly**	
	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP ($T = 0.5$)	68.52	29.93	93.17	0.66	16.35	13.01	57.93	41.47	71.26	15.60
MSP ($T = 0.75$)	68.59	29.93	93.17	0.66	16.36	13.01	57.95	41.47	71.38	15.45
MSP ($T = 1$)	68.65	29.94	93.18	0.66	16.37	13.00	57.95	41.47	71.46	15.41
MSP ($T = 1.1$)	68.67	29.94	93.18	0.66	16.37	13.00	57.95	41.51	71.47	15.40
MSP ($T = 5$)	69.06	29.95	93.18	0.66	16.42	13.00	57.95	41.58	71.85	15.33

5. Conclusion

The work shows how the effectiveness of anomaly segmentation depends not only on the architecture used, but mostly on the quality and specificity of the representations learned during pre-training. The best configuration is EoMT Cityscapes pre-trained model with softmax-based scoring, which combines highest mIoU with the most stable anomaly detection. Its main limitation remains the separation of true OoD objects from known classes, in particular the smallest ones (FS L&F). A natural improvement would be to combine post-hoc scoring with training dedicated to OoD detection.

6. Acknowledgments

Generative AI tools were not used for writing the project report. For the development of the submitted project code, Generative AI tools were utilized as assistive instruments.

GenAI systems used: Claude Sonnet 4.6 (Anthropic), Claude Haiku 4.5 (Anthropic), and Gemini Pro 3.1 (Google).

The tools were used during the development and experimental phases. Initially, they assisted in understanding the preliminary architectural structure of the project. Throughout development, it was used for target debugging. In particular, during Step 5 (fine-tuning), the tools provided baseline recommendations for hyperparameter starting points, which we then used to guide the tuning tasks. Finally, it was leveraged for code refactoring, cleaning, and final syntactic checks to optimize development time before submission.

Human supervision was continuous and strict engineering judgment was maintained at every stage. No AI output was accepted passively. Every suggestion, architectural draft, and debugging solution was critically analyzed, verified, and explicitly approved by at least two group members (according to our internal task division) before being integrated into the final codebase.

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